CYCLON convergence: proved cases and progress toward a general theorem

Research working note

19 October 2026

Abstract

We collect the current results in one self-contained source. The main new positive result proves convergence of the complete cache-and-age process for mandatory-removal CYCLON with three peers, cache capacity two, and shuffle length two, from every weakly connected initial state with arbitrary finite ages. A uniform regeneration argument gives an explicit total-variation bound and asymptotically uniform local samples conditional on availability. For arbitrary network size and cache capacity, we also prove a finite-set age-reset lemma when the shuffle length is one. A controlled replacement lemma treats general shuffle lengths but does not yet supply a complete forgetting argument.

Earlier results are retained: a capacity-one retention theorem and counterexamples for particular scheduling and retention conventions. Each statement specifies its assumptions. In particular, the new positive theorems use independent uniform activations rather than the fixed periodic phases in the original paper. A convergence proof for arbitrary network size, capacity, and shuffle length remains open here.

1 Scope and exact protocol conventions

The motivating target is a local total-variation guarantee of the kind proved for PEERSWAP in [1]. Its interchange-process reduction uses a fixed underlying graph and does not automatically extend to partial cache exchanges.

We consider a static membership $V = \{1, \dots, n\}$, reliable atomic exchanges, and caches without self entries or duplicate identities. Descriptors contain an identity and a nonnegative integer age. Enhanced CYCLON increments the initiator's cached ages, selects an oldest descriptor as partner, sends a fresh self descriptor in place of that partner along with other selected descriptors, and receives a subset of the partner's cache [2].

The update convention matters. We distinguish:

- *Retention:* discard self and already-known incoming descriptors, and replace sent cache entries only when a usable incoming descriptor is inserted. A contacted partner is retained when no replacement is usable. Discarded duplicates do not refresh ages.
- *Mandatory removal:* unconditionally remove the contacted partner, even when the response yields no usable replacement.

These conventions differ on reciprocal exchanges. The published paper's seven steps are followed by the explicit statement that the contacted peer disappears [2, Section 2.2]. We therefore use mandatory removal as the baseline for the new positive results. The retention results below concern the separately specified alternative; their obstructions do not establish failure of mandatory-removal CYCLON. Underfull buffers, replacement choices, and scheduling are specified explicitly wherever needed.

Time counts global activation events. Periodic phases and independent uniform random activations are treated separately. For a cache that can supply b entries, $S_i^{(b)}$ is a uniformly

selected, uniformly ordered b-tuple of distinct cached identities. When a requested tuple is unavailable, write $\perp$. Uniformity at one peer and time does not imply independence across peers or times.

2 A periodic counterexample under either convention

At $c = \ell = 1$, if the initiator is p, its neighbor is q, and q's cached identity is r, with p, q, r distinct, the unique exchange outcome is

$$(f(p), f(q)) = (q, r) \quad \longmapsto \quad (f'(p), f'(q)) = (r, p). \tag{1}$$

All other identities stay fixed. Both update conventions agree here.

Theorem 1 (Failure with fixed periodic phases)**.** *There is a four-peer enhanced-CYCLON execution with a strongly connected initial overlay, full caches, $c = \ell = 1$, and one activation per peer per round, in which a peer's sample never converges to uniform. No exchange has a tie, unusable replacement, or empty cache.*

Proof. Activate peers in order $1, 2, 3, 4$ in every round, using distinct fixed phases within a common period. Initialize all descriptor ages to zero and the identity state to $(3, 4, 2, 1)$. Equation (1) gives

After activation	$f(1)$	$f(2)$	$f(3)$	$f(4)$
Initial	3	4	2	1
1	2	4	1	1
2	2	1	1	2
3	3	1	2	2
4	3	4	2	1

The initial graph is the directed cycle $1 \to 3 \to 2 \to 4 \to 1$. In every exchange p, q, r are distinct. The initiator's sole descriptor is incremented from zero to one, then discarded and replaced in its outgoing message by its own age-zero descriptor. The responder's age-zero descriptor moves to the initiator. Thus all stored ages are again zero after each exchange. The entire state and scheduler phase repeat after four activations.

At each round boundary, peer 1 returns peer 3 deterministically. For all $k \geq 0$,

$$\left\| \mathcal{L}(S_1^{(1)}(4k)) - \mathrm{Unif}\{2, 3, 4\} \right\|_{\mathrm{TV}} = \frac{2}{3}.$$

Moreover peer 1 never stores peer 4, even between round boundaries. Every mixture over observation times therefore remains at least $1/3$ from uniform. Time averaging cannot repair this execution. □

3 Frozen multi-entry states with random scheduling

This section uses retention and discards already-known descriptors *without refreshing their ages.* It allows random initiators, random subsets, and any tie-breaking rule.

Theorem 2 (Strongly connected frozen states)**.** *For every even $n \geq 4$, every $2 \leq c \leq n - 2$, and every $1 \leq \ell \leq c$, there is a strongly connected, full-cache initial state for which all cache identities remain fixed forever under enhanced CYCLON with retention. Every peer always has a unique oldest descriptor. This holds for every activation sequence and every outcome of the protocol's random subset selections.*

For every $1 \le b \le c$ and every observer i, its sample distance is the positive constant

$$\left\| \mathcal{L}(S_i^{(b)}(t)) - \mathrm{Unif}((V \setminus \{i\})_b) \right\|_{\mathrm{TV}} = 1 - \frac{(c)_b}{(n-1)_b}, \qquad t \ge 0, \tag{2}$$

where $(a)_b = a(a-1)\cdots(a-b+1)$.

Proof. Partition peers into pairs (p_k, q_k), $1 \le k \le m = n/2$. For each pair choose a set S_k of $c-1$ identities outside the pair that contains p_{k+1}, with indices modulo m. This is possible since $m \ge 2$ and $c - 1 \le n - 3$. Set

$$C_{p_k} = \{q_k\} \cup S_k, \qquad C_{q_k} = \{p_k\} \cup S_k.$$

Give the reciprocal partner descriptor age 1 and every other descriptor age 0. Each pair is internally mutually reachable; the edges to p_{k+1} connect the pairs around a directed ring. Consequently the overlay is strongly connected.

Suppose p_k initiates. Its unique oldest descriptor is q_k. The outgoing message consists of its own fresh identity p_k and $\ell - 1$ elements of S_k, all already in q_k's cache. The responder therefore discards every incoming descriptor. Every descriptor it returns is either p_k, which is a self entry at the initiator, or an element of S_k, already known there. The initiator also discards every incoming descriptor. No cache identity and no stored age is overwritten.

Incrementing all ages at the initiating cache preserves its strict age difference between the partner and the common neighbors. Passive responses do not increment ages. The same argument applies when q_k initiates. Induction proves that the identities remain frozen and the designated partners remain uniquely oldest for all time.

The output is uniform on the $(c)_b$ ordered tuples contained in the fixed cache, whereas the target is uniform on all $(n-1)_b$ tuples. For uniform laws on a subset A of a finite set B, their total-variation distance is $1 - |A|/|B|$, proving (2). □

The smallest example, with $n = 4, c = 2$, is

$$C_1 = \{2^{[1]}, 3^{[0]}\}, \quad C_2 = \{1^{[1]}, 3^{[0]}\},$$
$$C_3 = \{4^{[1]}, 1^{[0]}\}, \quad C_4 = \{3^{[1]}, 1^{[0]}\},$$

where brackets denote ages. It works for both $\ell = 1$ and $\ell = 2$. Peer 1 never samples peer 4; its singleton distance is $1/3$. For a realistic illustration, $c = 20, \ell = 5$ is covered whenever even $n \ge 22$.

This counterexample uses a prescribed nonzero-age initialization. It does not apply to mandatory partner removal or duplicate-processing rules that refresh stored ages. The next proposition addresses reachability from zero ages for one parameter choice, while carefully distinguishing pathwise and distributional statements.

Proposition 3 (A frozen state reachable from zero ages)**.** *Under retention, $n = 4, c = 2, \ell = 1$, independent uniform initiators, uniform age-tie resolution, and uniform responder subsets, there is a strongly connected full-cache, zero-age initialization from which a frozen identity state is reached with probability at least 2^{-24}. Consequently almost-sure time-averaged uniform sampling fails for that initialization.*

Proof. Start from

$$C_1 = \{2^0, 4^0\}, \quad C_2 = \{3^0, 4^0\}, \quad C_3 = \{1^0, 2^0\}, \quad C_4 = \{1^0, 2^0\}.$$

Superscripts denote ages. This graph is strongly connected. Execute the following triples of initiator, selected partner, and returned identity:

$$(2,4,1),\ (1,4,2),\ (3,1,2),\ (3,1,4),\ (4,2,1),\ (4,2,3).$$

For $\ell = 1$, an accepted incoming descriptor replaces the single sent entry; a filtered descriptor makes no replacement. Direct substitution in these rules gives

$$C_1 = \{4^1, 3^0\}, \quad C_2 = \{3^1, 4^0\},$$
$$C_3 = \{2^2, 4^1\}, \quad C_4 = \{1^2, 3^1\}.$$

The pairs $(1, 4)$ and $(2, 3)$ now satisfy the frozen-state condition, with common sets $\{3\}$ and $\{4\}$ respectively. Each of the six specified initiators has probability $1/4$, each specified oldest-partner choice is between two tied entries, and each specified response has probability $1/2$. The exact prefix probability is therefore $(1/16)^6 = 2^{-24}$. On this event, peer 1 never again samples peer 2, so its empirical frequency for that identity tends to zero rather than $1/3$. □

Positive-probability absorption by itself does *not* prove that the unconditional local marginal law fails to approach uniform from the zero-age initialization: different frozen outcomes could balance in their mixture. Theorem 2 directly disproves that marginal-law guarantee when the specified nonzero initial ages are allowed.

4 An availability obstruction under mandatory removal

Now choose initiators independently and uniformly, remove the partner unconditionally, and let an activation of an empty cache do nothing.

Proposition 4 (Persistent failure of full-capacity requests)**.** *Let $c = n - 1$ and $1 \le \ell \le c$. For every peer i and $t \ge 1$, even starting from complete caches,*

$$\Pr\{S_i^{(c)}(t) = \perp\} \ge \frac{1}{n}, \qquad \left\| \mathcal{L}(S_i^{(c)}(t)) - \mathrm{Unif}((V \setminus \{i\})_c) \right\|_{\mathrm{TV}} \ge \frac{1}{n}.$$

Proof. If activation t selects i, either its cache is already empty or it removes some partner q. Since q's cache contains no self entry q, its response cannot restore q to i. Thus i has at most $n-2 < c$ entries at the end of that activation. The event that i was selected has probability $1/n$. The distance bound follows by testing the event $\{\perp\}$, which has target probability zero. □

This concerns unconditional $b = c$ requests. It does not establish nonuniformity for smaller samples conditional on availability. Retrying or conditioning changes the sampling model.

5 A positive capacity-one theorem

5.1 Precisely specified variant

Set $c = \ell = 1$, retain the current neighbor when a reply is unusable, and choose the initiator independently and uniformly at each step. With $q = f(p)$ and $r = f(q)$, the complete identity update is

$$T_p f = \begin{cases} f, & r = p, \\ f \text{ with } f(p) \leftarrow r,\ f(q) \leftarrow p, & r \neq p. \end{cases} \tag{3}$$

Initial ages may be arbitrary: the single cached entry is necessarily oldest, so ages have no effect on this identity projection. Let Ω_n be the nonself functions $f : V \to V$ whose underlying undirected graph is connected, and let

$$P(f, g) = \frac{1}{n} \#\{p : T_p f = g\}.$$

Theorem 5 (Convergence of the capacity-one retention variant)**.** *For every $n \geq 2$, $f \in \Omega_n$, and observer i,*

$$\mathcal{L}_f(f_t(i)) \longrightarrow \mathrm{Unif}(V \setminus \{i\}).$$

For $n \geq 4$, with $D = 2n(n-2)$ and $\delta = n^{-D}$,

$$\|\mathcal{L}_f(f_t(i)) - \mathrm{Unif}(V \setminus \{i\})\|_{\mathrm{TV}} \leq (1-\delta)^{\lfloor t/D \rfloor}. \tag{4}$$

Consequently $D\lceil n^D \log(1/\varepsilon)\rceil$ activations suffice for distance at most ε. For $n = 3$, $(8/9)^{\lfloor t/2 \rfloor}$ is sufficient. For $n = 2$, the sample is already uniform.

This is an extremely weak quantitative bound and gives no efficient mixing claim. The theorem concerns identity marginals, not stationarity of the complete age process.

5.2 Proof for $n \geq 4$

Lemma 6 (Inverse moves and connectivity)**.** *Ω_n is closed under T_p. Every nontrivial move initiated by p with $q = f(p)$ is inverted by initiating q in the resulting state. In particular P is symmetric.*

Proof. A nontrivial move replaces a path on the three distinct vertices p, q, r by another path on the same vertices, preserving undirected connectivity and avoiding self entries. In the resulting state $g(q) = p, g(p) = r$; initiating q restores $f(q) = r, f(p) = q$. This pairs each marked transition (f, p, g) with its unique reverse marked transition (g, q, f), so the transition counts agree. □

Lemma 7 (Insertion into a cycle)**.** *Every state in Ω_n reaches a directed Hamiltonian cycle in at most $n-2$ moves.*

Proof. A connected functional digraph has exactly one directed cycle: outgoing paths eventually reach a cycle, and an edge cannot join basins of distinct cycles. The cycle length is at least two. While an off-cycle vertex remains, choose p whose parent $q = f(p)$ lies on the cycle. Initiating p replaces its cycle edge $q \to r$ by $q \to p \to r$, adding one vertex to the cycle. □

Lemma 8 (Adjacent swap)**.** *If a Hamiltonian cycle has consecutive distinct vertices x, a, b, y, the activation word x, x, a, b swaps a, b, leaving every other outgoing edge unchanged.*

Proof. The only changing outgoing edges are

	$f(x)$	$f(a)$	$f(b)$
initial	a	b	y
x	b	x	y
x, x	y	x	x
x, x, a	a	y	x
x, x, a, b	b	y	a

All moves are valid for $n \geq 4$. At $n = 4$ the temporary residual cycle $x \leftrightarrow y$ is allowed. □

Lemma 9 (Diameter)**.** *For $n \geq 4$, the transition graph on Ω_n is connected with diameter at most $D = 2n(n-2)$.*

Proof. Reach a Hamiltonian cycle from each of two states in at most $n-2$ steps. Anchor the cyclic notation at one fixed vertex. The other $n-1$ vertices form a linear ordering; any two orders are related by at most $(n-1)(n-2)/2$ adjacent swaps. Each costs four moves by Lemma 8. Reverse the second insertion path using Lemma 6. The total is at most $2(n-2)+2(n-1)(n-2) = 2n(n-2)$. □

Proof of Theorem 5 for $n \geq 4$. The uniform law π_n on Ω_n is stationary because P is symmetric. Choose the state $z(1) = 2, z(2) = 1, z(j) = 1$ for $j \geq 3$. Initiating 1 at z is a self-loop. Every state reaches z along a path of at most D moves. Pad shorter paths with that self-loop to exactly D activations. The prescribed sequence has probability $n^{-D} = \delta$, so

$$P^D(f, \cdot) \geq \delta\, \delta_z(\cdot) \quad (f \in \Omega_n).$$

Writing $P^D(f, \cdot) = \delta\, \delta_z + (1 - \delta)R(f, \cdot)$ yields

$$\|\mu P^D - \nu P^D\|_{\mathrm{TV}} \leq (1 - \delta)\|\mu - \nu\|_{\mathrm{TV}}$$

for any probability laws μ, ν. Iterate with $\nu = \pi_n$, then use total-variation contraction for the remaining steps:

$$\|P^t(f, \cdot) - \pi_n\|_{\mathrm{TV}} \leq (1 - \delta)^{\lfloor t/D \rfloor}.$$

Relabelling bijects Ω_n onto itself and preserves π_n. Permutations fixing i act transitively on the other identities, so $\pi_n\{f : f(i) = j\} = 1/(n - 1)$ for each $j \neq i$. Projection gives (4). The time bound follows from $(1 - \delta)^k \leq e^{-\delta k}$. □

5.3 Small network sizes

For $n = 2$ the sole state is the reciprocal pair and its local target distribution has one element. For $n = 3$ the identity chain has two four-state classes. One consists of

$$H = (2, 3, 1), \quad L_1 = (3, 1, 1), \quad L_2 = (2, 1, 2), \quad L_3 = (3, 3, 2).$$

The other is obtained by relabelling a reversed triangle. The center H moves to each leaf with probability $1/3$. Each leaf returns to H with probability $1/3$ and stays with probability $2/3$. These classes contain all eight nonself maps. Uniform law on each class is stationary, and every observer points to either other identity in exactly two of its four states.

Choose a leaf $L_\star$ in the appropriate class. From every state, $P^2(f, L_\star) \geq 1/9$: from another leaf go through H; from H use the target leaf's self-loop; from the target leaf use two self-loops. The same minorization argument proves the stated bound. This also shows why irreducibility of the full identity space is stronger than necessary for local uniformity.

6 A general symmetry reduction, not a mixing proof

Lemma 10 (Symmetrized reference process)**.** *Let a Markov transition kernel commute with peer relabelling, and suppose the sample extraction rule is equivariant and always defined. Fix initial state x, observer i, and the group H_i of permutations fixing i. Start a reference process Y_t from*

$$\bar{\mu}_x = \frac{1}{|H_i|} \sum_{\sigma \in H_i} \delta_{\sigma x}.$$

Then $S_i^{(b)}(Y_t)$ is exactly uniform on the ordered distinct b-tuples of other peers at every time. For any coupling,

$$\|\mathcal{L}_x(S_i^{(b)}(X_t)) - \mathrm{Unif}((V \setminus \{i\})_b)\|_{\mathrm{TV}} \leq \Pr\{S_i^{(b)}(X_t) \neq S_i^{(b)}(Y_t)\}.$$

Proof. The initial law of Y is invariant under H_i. Equivariance preserves this invariance at every time and transfers it to the sample law. Since H_i acts transitively on the output tuples, that law must be uniform. The remaining claim is the coupling inequality. □

This lemma requires no finite state space, stationary age law, or full-state irreducibility. If samples can be unavailable, the uniformity conclusion holds only conditional on availability. The lemma provides no forgetting bound by itself. In Theorem 2, its reference law is uniform while the original local law remains permanently nonuniform, so the required disagreement probability cannot tend to zero.

7 Exact model for the mandatory-removal results

The positive results that follow use the enhanced exchange in the published CYCLON paper [2, Section 2.2], with the following explicit completion of its choices. Membership is fixed, exchanges are reliable and atomic, and $1 \le \ell \le c \le n-1$. Each cache C_p contains at most c distinct nonself identities, each with a finite nonnegative integer age.

At each global event, choose an initiator independently and uniformly from V. An empty initiator does nothing. Otherwise, the initiator p increments all its cached ages and selects a maximum-age partner q, breaking ties uniformly. It selects $\min(\ell-1, |C_p|-1)$ other entries uniformly. Write S for these other entries with their incremented ages. Its outgoing buffer is $\{p^0\} \cup S$, and it unconditionally removes q from its cache. The responder chooses a uniform subset of size $\min(\ell, |C_q|)$ of its *pre-exchange* cache and returns it. Both sides discard incoming self entries and incoming identities already present locally, without refreshing duplicate ages. They insert every remaining incoming entry, using vacancies first and then replacing sent entries still resident locally. Every eligible choice of victims has positive probability, independent of identities. For uniform quantitative bounds one may choose uniformly among the eligible sets. Entries not selected as victims remain in place.

The two buffers are formed before either cache processes incoming entries. Removing q creates a vacancy at p; it does not remove the other sent entries. At q, all sent entries are initially resident. These rules always provide enough eligible slots. If the pre-exchange cache sizes are a, b, the initiator has $c - a + 1 + \min(\ell-1, a-1) \ge \ell$ vacancies plus eligible residents, and the responder has $c - b + \min(\ell, b) \ge \ell$. The incoming buffer size at either side is at most ℓ.

This defines a countable-state Markov chain including ages. Independent equal-rate Poisson activation clocks give this same embedded chain. Fixed periodic phases, which the original paper uses, give a different scheduler. Thus none of the positive results below silently asserts convergence for every periodic schedule. The underfull-buffer convention and uniform tie-breaking above are also mathematical specifications, rather than claims that the paper uniquely prescribes them.

8 From full-state convergence to local uniformity

The following facts isolate what a general proof must establish. A Markov kernel is *equivariant* if relabelling peers in a state and then taking a step has the same law as taking a step and relabelling its outcome.

Lemma 11 (A common atom gives convergence, including ages)**.** *Let P be a Markov kernel on a countable state space. Suppose there are $L \ge 1$, $\delta > 0$, and a state z such that*

$$P^L(x, \{z\}) \ge \delta \qquad \text{for every state } x.$$

Then P has a unique stationary probability law π, and

$$\sup_x \|P^t(x,\cdot) - \pi\|_{\mathrm{TV}} \le (1-\delta)^{\lfloor t/L \rfloor}.$$

Irreducibility of the whole state space is not required.

Proof. The case $\delta = 1$ is immediate. Otherwise split $P^L(x,\cdot) = \delta\delta_z + (1-\delta)R(x,\cdot)$, where R is a Markov kernel. Hence $\mu \mapsto \mu P^L$ is a contraction by $1-\delta$ in total variation. Probability measures on a countable space are complete in this metric, so the contraction has a unique fixed point π. Since $(\pi P)P^L = (\pi P^L)P = \pi P$, uniqueness gives $\pi P = \pi$. Every P-stationary law is also P^L-stationary, proving uniqueness for P. Iterating the contraction and applying the remaining $t \bmod L$ transitions proves the bound. □

Theorem 12 (Stationary symmetry and availability)**.** *Let P be equivariant on a relabelling-invariant state space. Suppose it has a unique stationary law π and $\eta_t(x) = \|P^t(x,\cdot) - \pi\|_{\mathrm{TV}} \to 0$. Fix i and $1 \le b \le c$, and suppose $a = \pi\{|C_i| \ge b\} > 0$. For uniform sampling without replacement from C_i, followed by uniform ordering,*

$$\mathcal{L}_\pi(S_i^{(b)} \mid |C_i| \ge b) = \mathrm{Unif}((V \setminus \{i\})_b).$$

The same conditional sample law converges to this uniform law from every initial state x. If $\eta_t(x) \le a/2$, its distance is at most $4\eta_t(x)/a$.

Proof. For every permutation σ of V, equivariance makes $\sigma_*\pi$ stationary. Uniqueness gives $\sigma_*\pi = \pi$. The event $A = \{|C_i| \ge b\}$ is invariant under permutations fixing i, and the output rule is equivariant under this group. Its action on distinct ordered b-tuples of other identities is transitive. An invariant probability law on that finite set is uniform.

For the quantitative claim include the extra randomness of sample extraction in the transition from a state to $(A, S_i^{(b)})$. Total variation cannot increase under this transition. Put $a_t = \Pr_x(A \text{ at } t)$. Then $|a_t - a| \le \eta_t(x)$. For any set B of available tuples,

$$\left| \frac{\Pr_x(A, S_i^{(b)} \in B)}{a_t} - \frac{\Pr_\pi(A, S_i^{(b)} \in B)}{a} \right| \le \frac{2\eta_t(x)}{a_t} \le \frac{4\eta_t(x)}{a}.$$

Taking the supremum over B proves the result. □

At stationarity the *unconditional* output law, including the unavailable symbol, is $(1-a)\delta_\perp + a\,\mathrm{Unif}((V \setminus \{i\})_b)$. Its distance from the always-available uniform target is exactly $1-a$. Conditioning at a fixed observation time is the sampling rule in Theorem 12. Waiting or retrying until a cache can answer introduces a stopping-time rule and requires a separate argument. Nor does this theorem assert independent samples across peers or time, or a uniform distribution on global overlay graphs.

9 A general age-reset result for shuffle length one

Theorem 13 (Finite-set regeneration for $\ell = 1$)**.** *Use the mandatory-removal model of Section 7 with arbitrary n, c and $\ell = 1$. Label the peers $1, \dots, n$ and set*

$$K = n + c - 1, \qquad L = nK, \qquad \delta = (nc)^{-L}.$$

Let $\mathcal{F}$ be the finite set of states with all ages zero and $C_i \subseteq \{i+1, \dots, n\}$ for every i. Then, from every state with arbitrary finite ages,

$$P^L(x, \mathcal{F}) \ge \delta.$$

If $\tau_{\mathcal{F}}$ is its first hitting time, then

$$\Pr_x\{\tau_{\mathcal{F}} > mL\} \le (1-\delta)^m, \qquad \mathbb{E}_x \tau_{\mathcal{F}} \le L/\delta.$$

Proof. First activate one fixed peer p for K consecutive events. Only its cache ages are incremented during this block. When its chosen partner q already knows p, prescribe that q return the descriptor for p. This occurs with probability at least $1/c$. The reply is a self entry at p, so mandatory removal reduces $|C_p|$ by one. The request contains only p^0, a duplicate at q; hence q retains its descriptor for p.

If q does not know p, its receipt of p^0 makes it learn p. It either uses a vacancy or replaces its single returned entry. Subsequent exchanges in this block cannot make it forget p: it never

initiates, and every request from p contains only the already-known identity p. There are at most $n-1$ such first-learning events. The size of C_p never increases, because one descriptor is removed and at most one is received. Every other nonempty activation in the prescribed block shrinks it. At most c shrinkings empty it. Consequently $n-1+c=K$ activations suffice, independently of all ages. Remaining activations of an empty p do nothing.

Now execute this block for $p=1,2,\dots,n$ in order. Immediately after its block, p is empty. During later blocks it never initiates, and every incoming request contains only a fresh self descriptor j^0 from a later peer $j>p$. Thus its final cache consists solely of age-zero descriptors of later peers. The final state belongs to $\mathcal{F}$.

Prescribing the L initiators has probability n^{-L}. Each reply that must be prescribed has conditional probability at least $1/c$; there are at most L of them. No particular oldest-partner or victim choice was required. The resulting lower bound is $(nc)^{-L}$. Applying it to successive L-event blocks, using the Markov property, gives the tail bound, and summing that bound gives the expectation. □

This proves that unbounded initial ages can be forgotten often enough to hit a finite family. It does not prove that the different members of $\mathcal{F}$ communicate or have identical limiting local marginals. In particular, $\mathcal{F}$ need not have a unique empty sink. Weak connectivity is preserved for $\ell=1$: the contacted edge is retained in reverse, and if the responder deletes its sole returned edge, that edge is transferred to the initiator or is already present there. Hence the same hitting statement holds for $\mathcal{F}$ intersected with the weakly connected states when that is the initial state space. A common-atom construction within this finite family is still missing for general n, c.

10 A complete multi-entry mandatory-removal theorem

Theorem 14 (Three peers with two cache entries)**.** *In the mandatory-removal model of Section 7, let $n=3$ and $c=\ell=2$. Start from any weakly connected identity overlay, with arbitrary finite nonnegative integer ages. The complete cache-and-age chain has a unique stationary law π on this state space, and*

$$\sup_x \|P^t(x,\cdot)-\pi\|_{\mathrm{TV}} \le (1-6^{-20})^{\lfloor t/20\rfloor}. \tag{5}$$

For every observer and $b=1,2$, the stationary probability of availability is positive and the conditional local sample converges to $\mathrm{Unif}((V\setminus\{i\})_b)$.

The bound is deliberately crude. This is a proof for actual oldest-first, age-carrying exchanges with multi-entry caches, but it does not establish a theorem for arbitrary n, c, ℓ. At $n=3$ the conditional $b=2$ output is automatically uniform when both possible neighbors are present; convergence and positive availability still require proof.

10.1 Three-peer transition rules

Write q^a for a descriptor of identity q and age a. A full cache contains both possible other identities. The responder never needs to delete an entry: it simply adds any new nonself identity, since only two such identities exist. A full initiator ends with exactly its other, uncontacted entry, whose age was incremented. It cannot receive the contacted identity from that identity's own cache. A singleton initiator ends with the responder's third identity, if present, and is otherwise empty.

The weakly connected state space is closed. If a full cache is present after an exchange, all three vertices are connected. Otherwise an exchange with an empty responder reverses an edge, and a reciprocal singleton exchange deletes only one of the two reciprocal arcs. A path through distinct singleton initiator and responder creates a full responder. These cases cover every exchange and preserve weak connectivity.

10.2 Resetting arbitrary ages on a chain

For three distinct role names F, S, T, call the state a *chain* if

$$C_F = \{S^a, T^b\}, \qquad C_S = \{F^d\}, \qquad C_T = \{S^e\}. \tag{6}$$

Define the zero-age state $Z(F, S, E)$ by

$$C_F = \{S^0, E^0\}, \qquad C_S = \{E^0\}, \qquad C_E = \varnothing.$$

Lemma 15 (Chain age reset)**.** *Every chain reaches a state of form Z in at most eight exchanges, regardless of its finite initial ages.*

Proof. Suppose first $b \geq a$. Prescribe the following five exchanges:

$$F \to T, \quad S \to F, \quad F \to S, \quad T \to S, \quad T \to F.$$

The complete trace is

after	C_F	C_S	C_T
initial	$\{S^a, T^b\}$	$\{F^d\}$	$\{S^e\}$
$F \to T$	$\{S^{a+1}\}$	$\{F^d\}$	$\{S^e, F^0\}$
$S \to F$	$\{S^{a+1}\}$	$\varnothing$	$\{S^e, F^0\}$
$F \to S$	$\varnothing$	$\{F^0\}$	$\{S^e, F^0\}$
$T \to S$	$\varnothing$	$\{F^0, T^0\}$	$\{F^1\}$
$T \to F$	$\{T^0\}$	$\{F^0, T^0\}$	$\varnothing$

The fourth choice is legal because the two ages at T after its increment are $e+1$ and 1; S is oldest, possibly tied when $e = 0$. All other partner choices are forced by singleton caches, apart from the first, which is allowed by $b \geq a$. The final state is $Z(S, F, T)$.

If instead $a > b$, activate F toward S. This produces a chain with roles (S, T, F) and full-cache ages $(b+1, d)$ for its near and far vertices, respectively. If $d \geq b+1$, the preceding reset applies. Otherwise integrality gives $d \leq b$; rotate again by activating the new full vertex toward its near vertex. The roles are now (T, F, S) and the full ages are $(d+1, e)$. If $e \geq d+1$, reset. Otherwise $e \leq d \leq b$; a third rotation gives roles (F, S, T) with full ages $(e+1, b+1)$. Now the far vertex is oldest, and the five-exchange reset applies. At most three rotations and five further exchanges are needed. □

10.3 Reaching the chain topology

Lemma 16 (Identity reduction compatible with oldest-first choices)**.** *Every weakly connected three-peer state reaches a chain in at most seven exchanges.*

Proof. Classify states by the numbers of full, singleton, and empty caches. All named activations below use an oldest entry; where it matters, the age inequalities are given explicitly.

Two full caches and one singleton. Let A, B be full and C singleton. Activate A. If it contacts B, there is now one full cache and two singletons. Otherwise it contacts C, which becomes full, and A becomes singleton. If C previously pointed to B, its full cache has ages B^u, A^0 for some $u \geq 0$. Activate C toward the full vertex B; this is oldest or tied and reduces to one full cache. If C previously pointed to A, its cache is $\{A^u, B^{v+1}\}$, where v was the age of B at A. If B is oldest, again activate C toward B. Otherwise activate C toward the singleton A. Then A is full with cache $\{B^{v+1}, C^0\}$. It can contact B, its strictly oldest entry. Thus at most three exchanges reach one full cache and two singletons.

Other states with at least two full caches. Three full caches become two full and one singleton after one activation. With two full and one empty, activate a full cache. It either contacts the

other full cache, yielding one full, one singleton, and one empty, or contacts the empty vertex, yielding two full caches and a singleton. Use the preceding case thereafter.

One full cache and two singletons. If the singleton arrows form a chain ending at the full vertex, the required form already holds. If both point to the full vertex, activate either singleton; it now points to the other singleton, giving a chain. The remaining case has reciprocal singleton arrows. Let Q be an oldest neighbor of the full vertex F and R the other vertex. Activate $Q \to R$, emptying Q, then $F \to Q$. The resulting state is a chain with full vertex Q. At most two exchanges are needed.

One full F, one singleton S, and one empty E. If $S \to F$, first activate it, giving $S \to E$. If the oldest neighbor of F is S, activate $S \to E$, emptying S; this does not change the ages at F. Now activate F toward an empty oldest neighbor. This gives one full cache and two singletons, handled above. There are at most three preliminary exchanges.

One full and two empty. One full activation gives the preceding pattern.

No full caches. With three singletons, weak connectivity implies a directed path $P \to Q \to R$ on distinct vertices. Activate $P \to Q$; Q becomes full and the other caches are singleton. With two singletons and one empty, there is either a path toward the empty root or two arrows into that root. In the path case, activate the leaf toward its singleton parent, producing one full, one singleton, and one empty. In the two-arrow case, reverse one arrow by activating it toward the empty root, then use the path case. This takes at most two preliminary exchanges. There are no other weakly connected size patterns.

The longest route just described is the last pattern: at most two exchanges to one full, one singleton, and one empty; at most three to one full and two singletons; and at most two to a chain. It costs seven. The two-full/empty route costs at most $1+3+2=6$, the three-full route at most $1+3+2=6$, and the other routes are no longer. □

10.4 A common zero-age state

Lemma 17 (Relabelling the zero-age roles by legal exchanges)**.** *Every $Z(F,S,E)$ reaches every prescribed labelled state of form Z in at most four exchanges.*

Proof. The exchange $S \to E$ maps $Z(F,S,E)$ to $Z(F,E,S)$ in one step. The two exchanges $F \to S$ followed by $F \to E$ map $Z(F,S,E)$ to $Z(S,E,F)$. The first choice is an age-zero tie; before the second exchange the remaining entry of F has age one. Direct substitution shows that both final states have all ages zero.

If the desired full vertex is currently S, use the two-step rotation. If it is currently E, first swap S, E and then rotate, costing three steps. If it is already F, do nothing. One final swap, if needed, sets the two remaining roles. □

Proof of Theorem 14. Lemmas 16 and 15 reach some zero-age Z in at most $7+8=15$ exchanges. Lemma 17 reaches the fixed state $z = Z(1,2,3)$ within four more. Pad the resulting path to exactly twenty activations by repeatedly activating the empty vertex at z. Every prescribed initiation has probability $1/3$ and every prescribed oldest choice has probability at least $1/2$. The buffers are deterministic in this parameter setting, and the responder never chooses victims. Therefore

$$P^{20}(x, \{z\}) \geq 6^{-20} \quad \text{for every weakly connected } x.$$

Lemma 11 proves existence, uniqueness, and (5), without truncating ages or assuming a finite state space.

The same construction works for any relabelled z. Integrating its minorization against π shows $\pi(\{z'\}) \geq 6^{-20}$ for each labelled Z state z'. For each observer choose one with that observer full. Thus $\pi\{|C_i| \geq b\} \geq 6^{-20} > 0$ for $b = 1, 2$. Theorem 12 gives the conditional local conclusion. □

11 Controlled progress for general shuffle lengths

The next lemma addresses one difficulty caused by larger buffers: copying a newly created descriptor can evict another such descriptor. It concerns a prescribed legal exchange, rather than the average behavior of the random process.

Attach an auxiliary color, old or fresh, to each descriptor. Copies inherit their source color, and every new age-zero self descriptor is fresh. Discarding a duplicate does not change the stored descriptor's color. Colors are proof bookkeeping and do not change the protocol or its oldest-first selection.

Lemma 18 (A controlled fresh-count inequality)**.** *Use Section 7 with arbitrary n, c, ℓ. Suppose a nonempty initiator p selects an old partner descriptor q. For every prescribed extra initiator buffer S, there exist a responder buffer and eligible victim choices for which the global number N_f of stored fresh descriptors does not decrease. It increases by at least one if the responder accepts any fresh incoming descriptor. All these choices have positive probability under uniform buffer and eligible-victim selection.*

Proof. Let $P = C_p$ and $Q = C_q$ denote the pre-exchange identity sets, $a = |P|$, $b = |Q|$, $k = |S| = \min(\ell - 1, a - 1)$, and $m = \min(\ell, b)$. Let S_f be the fresh descriptors in S, using their identities when taking intersections. The responder must insert

$$J = (\{p\} \cup S) \setminus Q, \qquad j = |J|.$$

Of these, $f_J = 1_{\{p \notin Q\}} + |S_f \setminus Q|$ are fresh. There are $v_Q = c - b$ vacancies, so $r_Q = (j - v_Q)_+$ entries must be replaced.

Call a descriptor in Q protected if it is fresh and its identity lies in $P \cup \{p\}$. Put $d_0 = |Q \cap (\{p\} \cup S)|$. Since $q \notin Q$, at most $d_0 + (a - 1 - k)$ descriptors are protected. Thus the number of nonprotected entries is at least

$$b - d_0 - a + 1 + k \geq k + 1 - d_0 - (c - b) = j - v_Q.$$

Choose r_Q nonprotected victims D. Choose the responder buffer T to include these victims and the entries with identities

$$B = (S_f \cap Q) \cup (\{p\} \cap Q).$$

This is possible: the union lies in Q, and if $r_Q > 0$ its size is at most $d_0 + r_Q = k + 1 - v_Q \leq \ell$. If $r_Q = 0$, its size is at most $k + 1 \leq \ell$. Complete T arbitrarily to size m. Fill vacancies and then replace exactly D. Every fresh victim at the responder has identity absent from $P \cup \{p\}$; its fresh copy in T is therefore accepted by the initiator.

The initiator accepts $I = T \setminus (P \cup \{p\})$. Write $i = |I|$, f_I for its fresh count, $f_S = |S_f|$, $d = |S_f \cap Q|$, and $h = 1_{\{p \in Q\}}$. Because T includes B, $i \leq m - d - h$. Deleting the old partner leaves $v_P = c - a + 1$ vacancies. Choose old extra-buffer victims before fresh ones. The number of fresh descriptors the initiator must remove is at most

$$\big(i - v_P - (k - f_S)\big)_+ \leq (f_S - d - h)_+ = (f_J - 1)_+.$$

Here $m - v_P \leq k$ follows from both $m - v_P \leq \ell - 1$ and $m - v_P \leq a - 1$. The responder loses at most f_I fresh descriptors, all of which have fresh copies accepted at the initiator. Combining the two endpoints gives

$$\Delta N_f \geq f_J - (f_J - 1)_+ = 1_{\{f_J > 0\}}.$$

This proves the controlled inequality and its strict version. □

This is neither a pathwise monotonicity theorem for every random exchange nor an expected-drift inequality. It does not handle an activation that removes a fresh partner. It also does not show that a strict increase can always be arranged within a bounded number of activations. That strict-progress problem remains a gap.

There is a limited bridge to numerical ages: over H global events, any descriptor lineage gains at most H age units, and copying preserves its age. Let A be the maximum of zero and all initial fresh-descriptor ages. If every initially old descriptor has age greater than $A + H$, an initiator containing an old descriptor selects an old partner throughout this horizon; new fresh descriptors start at age zero. A bounded colored clearing strategy compatible with actual oldest choices would therefore help construct an age reset. Such a strategy has not been proved here for arbitrary parameters.

12 Established results and remaining problem

The positive conclusions now cover two different settings. Theorem 5 proves local convergence for the specified capacity-one retention variant. Theorem 14 proves full-state convergence, including arbitrary finite initial ages, for mandatory-removal CYCLON with $n = 3, c = \ell = 2$ under independent uniform activations. Its proof explicitly combines topology control and an age reset, rather than assuming either.

For general n, c with $\ell = 1$, Theorem 13 proves a uniform hit on a finite zero-age family. What remains is to connect that family to a common recurrent regime, or to show that every possible limiting regime has the same uniform local marginals. For larger ℓ, Lemma 18 supplies controlled nondecrease of a freshness count, but a bounded strict-progress and topology argument is missing. The relevant state space must also be shown closed under the precise replacement rule before invoking a symmetry argument on that space.

A sufficient route to a general theorem is to construct, for each n, c, ℓ, a labelled reset state z and a bound $H(n, c, \ell)$ such that every admissible aged state has a legal reset path of length at most H, with a uniformly positive path probability and suitable padding to one common length. Lemma 11 would then prove full-state convergence; Theorem 12 would yield conditional uniform sampling. This sufficient condition is stronger than necessary: local convergence may hold without a unique full-state recurrent class.

The scheduling and sampling target cannot be omitted. Theorem 1 rules out a theorem covering every fixed periodic phase schedule at $c = \ell = 1$. The retention-specific frozen states in Theorem 2 do not refute the mandatory-removal target. Proposition 4 concerns unconditional full-capacity requests and does not refute conditional uniformity. The present note does not prove the arbitrary-parameter CYCLON convergence claim, nor any efficient or polylogarithmic mixing bound.

13 Finite verification

The accompanying `verify.py` checks the periodic trace including ages, every possible initiating/subset choice in a small frozen-twin example, the zero-age absorption certificate, the frozen construction's connectivity, the availability argument on small cache states, and the capacity-one identity chain exhaustively for $2 \le n \le 6$. It checks inverse moves, communicating classes, stationary local counts, and Hamiltonian adjacent swaps. It covers 14,831 connected identity states, 87,858 marked transitions, and 564 Hamiltonian swap words.

The additional checker `verify_reset.py`, using `cyclon3.py`, both in `mandatory-removal/`, follows the constructive three-peer reset without truncating the ages reached along its paths. It exhaustively checks all 46,550 weakly connected initial states whose stored ages belong to $\{0, 1, 2, 3, 4\}$. Every constructed transition is checked against the exact oldest-first kernel; every

path reaches $Z(1,2,3)$ within the proved nineteen-exchange limit. The longest observed path in this finite check has fourteen exchanges. The symbolic proof above, rather than this finite enumeration, establishes the theorem for arbitrarily large finite integer ages. All these checks corroborate the proofs; none replaces a missing general convergence argument.

The complete mathematical text is contained in this single `main.tex`; it has no external TeX inputs or bibliography file.

References

[1] R. Guerraoui, A.-M. Kermarrec, A. Kucherenko, R. Pinot, and M. de Vos. *PEERSWAP: A Peer-Sampler with Randomness Guarantees*. arXiv:2408.03829, 2024. https://arxiv.org/pdf/2408.03829.

[2] S. Voulgaris, D. Gavidia, and M. van Steen. *CYCLON: Inexpensive Membership Management for Unstructured P2P Overlays*. Journal of Network and Systems Management, 13:197–217, 2005. Published article, author-hosted copy: https://acropolis.aueb.gr/~spyros/www/papers/Cyclon.pdf.